Quantum Asset Pricing under Relativistic–Thermodynamic General Equilibrium

A Radial Framework for In-Planet, On-Planet, and Off-Planet Assets

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Abstract

Traditional asset-pricing theory explains financial valuation through systematic risk, stochastic discount factors, arbitrage, and intertemporal preferences. This paper proposes an additional state variable: the *radial origin* of productive assets. Every productive asset ultimately originates either within a planet, on its surface, or beyond the planet. These three classes are governed by distinct physical constraints. In-planet assets are constrained primarily by geology, thermodynamics, chemistry, and extraction technology. Surface assets are governed largely by conventional economic and institutional factors. Off-planet assets are constrained by relativity, communication delays, propulsion, radiation, and space logistics. We propose a radial framework for asset pricing in which equilibrium returns incorporate accessibility risk in addition to conventional systematic risk. The framework naturally extends modern portfolio theory, general equilibrium, welfare economics, and space finance.

The paper ends with "The End"

Keywords: Asset Pricing, Quantum Finance, General Equilibrium, Thermodynamics, Relativity, Space Economics, Portfolio Theory.

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1 Introduction

Modern financial economics has produced a wide range of successful valuation frameworks, including the Capital Asset Pricing Model (CAPM), Arbitrage Pricing Theory (APT), consumption-based asset pricing, stochastic discount factor models, and continuous-time pricing theory. These frameworks explain asset prices using systematic risk, intertemporal substitution, arbitrage, and market completeness [1, 2, 3, 4, 5, 6, 7, 8].

Despite their success, these theories generally abstract from the physical origin of productive assets. Every productive asset exists somewhere in physical space and must ultimately be accessed through geological, technological, or astronomical processes.

This observation motivates a simple extension.

Rather than classifying assets solely according to financial characteristics, we classify them according to their radial origin relative to the planetary surface.

2 Radial Classification of Assets

Let

$$\mathcal{I}$$

denote the set of in-planet assets,

$$\mathcal{S}$$

the set of surface assets,
and

$$\mathcal{E}$$

the set of off-planet assets.
Every asset belongs to one of these classes,

$$R_i \in \{\mathcal{I}, \mathcal{S}, \mathcal{E}\}.$$

Examples are summarized in Table 1.

Table 1: Radial classification of productive assets

Asset Class	Representative Examples
In-Planet	Mineral deposits, geothermal resources, petroleum, underground storage
Surface	Real estate, factories, agriculture, ports, infrastructure
Off-Planet	Satellites, lunar facilities, asteroid mining, orbital power stations

3 Accessibility as a Pricing Factor

The central hypothesis of this paper is that investors price not only uncertainty about future cash flows but also uncertainty regarding *physical accessibility.*

The traditional CAPM states

$$E(R_i) - r_f = \beta_i \lambda_M.$$

The proposed radial model becomes

$$E(R_i) - r_f = \beta_i \lambda_M + \gamma_i \lambda_A,$$

where

- λ_M is the conventional market risk premium,
- λ_A is the accessibility premium,

- γ_i measures exposure to accessibility risk.

Accessibility differs across radial sectors.

For geological assets it depends upon extraction technology, thermodynamics, pressure, chemistry, and depletion.

For surface assets it depends primarily upon transportation, institutions, political stability, and infrastructure.

For off-planet assets it depends upon launch capability, communication delay, propulsion, radiation, and orbital dynamics.

4 A Quantum-Inspired Representation

The term *quantum* is used here in a mathematical rather than a physical sense.

Represent each asset by the state vector

$$|\Psi_i\rangle = |R_i\rangle \otimes |F_i\rangle,$$

where

- $|R_i\rangle$ represents radial origin,
- $|F_i\rangle$ contains conventional financial characteristics.

Pricing becomes

$$\hat{P}|\Psi_i\rangle = P_i|\Psi_i\rangle.$$

This notation provides a compact representation for combining financial characteristics with physical accessibility.

5 Relativistic–Thermodynamic Constraints

The Relativistic–Thermodynamic General Equilibrium framework establishes that productive activity is governed by different physical constraints depending upon radial direction.

In-planet assets encounter increasing temperature, pressure, chemical hostility, and extraction costs.

Off-planet assets encounter finite signal speed, communication delays, propulsion requirements, radiation, and relativistic causal constraints.

Consequently, physical accessibility becomes an economically meaningful state variable.

6 Portfolio Theory

Portfolio diversification naturally extends to radial diversification.

Let

$$w_I + w_S + w_E = 1,$$

where

- w_I denotes the weight on in-planet assets,
- w_S denotes the weight on surface assets,
- w_E denotes the weight on off-planet assets.

The covariance matrix becomes

$$\Sigma = \begin{pmatrix} \Sigma_{II} & \Sigma_{IS} & \Sigma_{IE} \\ \Sigma_{SI} & \Sigma_{SS} & \Sigma_{SE} \\ \Sigma_{EI} & \Sigma_{ES} & \Sigma_{EE} \end{pmatrix}.$$

Diversification therefore extends naturally across radial sectors in addition to industries, countries, and currencies.

7 General Equilibrium

Within general equilibrium, labor, capital, technology, and research are allocated among the three radial sectors.

Investment in geological technology expands the economically accessible in-planet opportunity set.

Investment in propulsion, robotics, artificial intelligence, and autonomous systems expands the economically accessible off-planet opportunity set.

Thus asset pricing becomes linked to technological progress through physical accessibility.

8 Applications

Potential applications include

- deep geothermal finance,
- critical mineral investment,
- carbon sequestration,
- satellite infrastructure,
- orbital manufacturing,
- lunar development,
- asteroid mining,
- planetary-defense finance,
- space infrastructure investment.

As commercial space activity expands, physical accessibility may become an increasingly important determinant of asset valuation.

9 Future Research

Several extensions naturally follow.

First, accessibility factors may be incorporated into stochastic discount factor models.

Second, radial accessibility can be introduced into multi-factor asset pricing.

Third, reinforcement learning and artificial intelligence can be used to construct dynamic radial portfolios.

Finally, the framework can be integrated with Relativistic–Thermodynamic General Equilibrium to produce a unified theory of production, finance, and interplanetary economic development.

10 Conclusion

Traditional asset-pricing theory emphasizes uncertainty, preferences, and market structure.

This paper argues that the physical origin of productive assets should also enter financial valuation.

Three broad asset classes emerge:

1. in-planet assets,
2. surface assets,
3. off-planet assets.

Each class is governed by distinct physical constraints and therefore may carry distinct equilibrium accessibility premia.

Rather than replacing established financial theory, the proposed framework extends it by incorporating physical accessibility into the pricing of productive capital.

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Glossary

Accessibility Premium
Additional expected return associated with physical accessibility.

In-Planet Asset
A productive asset originating beneath the planetary surface.

Off-Planet Asset
A productive asset originating beyond the planetary surface.

Quantum State
A mathematical representation combining radial origin and financial characteristics.

Radial Asset Pricing
Asset pricing framework in which physical origin influences equilibrium valuation.

Radial Origin
The physical location of an asset relative to the planetary surface.

Surface Asset
A productive asset located on the planetary surface.

The End